# Palaeomagnetism of some Indian Rocks

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(Received 1962 May 30)

*Summary*

Hitherto, no palaeomagnetic data for India are available earlier than the Jurassic, 130 M.y. The remanent magnetic directions of oriented samples from one Jurassic and four pre-Cambrian rock formations in India have now been determined. One pre-Cambrian formation gave very scattered direction: the other four formations gave well-defined directions and their magnetic stability was demonstrated by a.c. and thermal demagnetization. The three pre-Cambrian results enable values of the ancient latitude and orientation of India to be estimated at 500, 600 and 750 M.y., assuming the ancient field to have been an axial dipole.

By making use of these new results, together with similar data by other workers for Australia, North America and Europe, it can be shown that, as far as the scanty data goes, the rate of drift in latitudes and orientation of the four continents has been of roughly the same magnitude throughout the period since 750 M.y.

## 1. Introduction

The earlier palaeomagnetic studies (Clegg & others 1956, 1958; Deutsch & others 1958, 1959) of the Deccan traps and the Rajmahal traps of India gave strong indications for the northward drift of the land mass of India during the last 150–200 M.y., assuming that the magnetic field has always been that of an axial dipole. The need for extending the study to other older igneous formations has become obvious. Unfortunately there is a singular lack of suitable igneous formations between the Cambrian and the Jurassic periods, and the few studies made on some of the sedimentary horizons have not thus far yielded any satisfactory results. However, attempts are still being made.

In the meanwhile, new palaeomagnetic studies have recently been made on the following igneous rock formations in India:

(a) Sylhet traps (Jurassic 120 to 150 M.y.)
(b) Malani Rhyolites (pre-Cambrian)
(c) Bijawar traps ,,
(d) Cuddapah traps ,,
(e) Mundwara complex ,,

The ages given for the above formations are mostly based on geological estimates (Dubey 1960) and conform to Holmes B time scales. Figure 1 is a map of India giving the new sites studied as well as those of the earlier work.

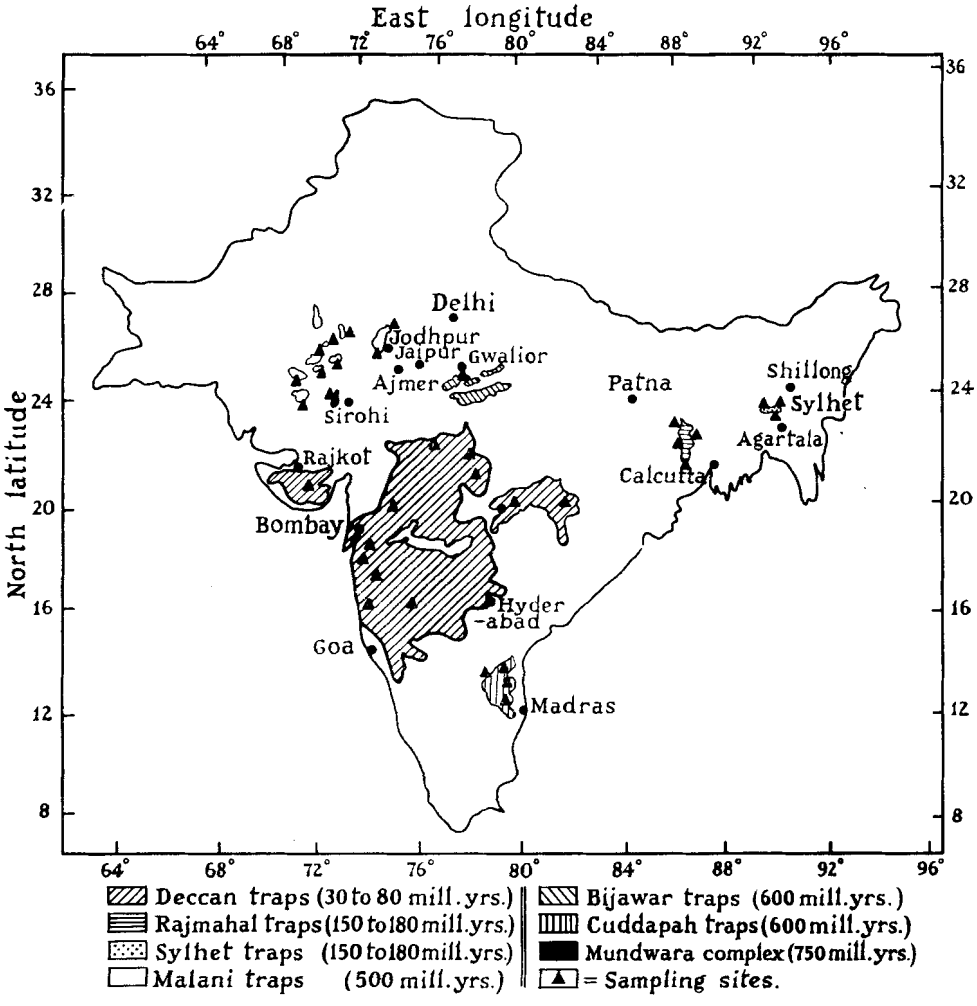



FIG. 1.—Detail of rock formations studied.

## 2. Geology of the formations studied

(a) *Sylhet traps.*—This formation (Medlicott 1865) consists of basaltic lava flows and associated dolerite dykes and is exposed along the southern base of Khasi hills in the united Khasi and Jaintia district of Assam. These traps occur overlying the Archaean crystalline metamorphic rocks and are themselves overlain by the Cretaceous Sandstones. The stratigraphic position of these traps has not been fixed definitely but they are generally regarded as contemporary to the Rajmahal traps of Bihar. Their age, therefore, has been taken to be Jurassic.

The extent of Sylhet traps exposure is itself small and even this is situated in thick forest region. A total of 36 oriented samples was collected from freshly made road cuttings covering a span of about 15 miles. Since the area covered by these traps is small and their age is believed to be Jurassic, the palaeomagnetic analysis of these traps is taken mostly as supplementary to the earlier studies on the Rajmahal traps of Bihar.

(b) *Malani Rhyolites.*—The Malani Rhyolites (La Touche 1902) cover thousands of square miles around Jodhpur in the state of Rajasthan and occur as isolated

hillocks. They are found lying undisturbed over the Aravali system of Huronian age and show a marked unconformity with the overlying upper Vindhyan Sandstones. These are generally grouped with lower Vindhyan rocks (Torridonian age). The flows in different Rhyolite outcrops are in general horizontal but occasionally appear to exhibit low angles of dip. In all, 60 oriented samples were collected from 9 different localities scores of miles apart.

(c) *Bijawar traps and* (d) *Cuddapah traps.*—Pre-Cambrian basic igneous rocks occur as dykes, sills and lava flows among sedimentary formations in the Cuddapah basin in Andhra Pradesh and round about Gwalior and Bijawar in Madhya Pradesh. These basic rocks occur in lower Cuddapah system (Wadia 1953) which is supposed to be of Algonkian age.

The Cuddapah basin is presumed to have been subjected to diastrophic movements of various kinds and the formations in it show tilts of varying magnitudes. The extent to which the Cuddapah traps have been disturbed cannot be easily ascertained. It is, however, likely that they have also been affected to the same extent as is indicated by the underlying and overlying sedimentary strata. The occurrence of slates and Quartzites in the Cuddapah formation indicates some metamorphism. However, it is generally considered that the traps have not been metamorphosed to any considerable extent. 72 oriented samples were studied from 10 different sites covering the whole of the Cuddapah trap area.

The traps occurring at Gwalior belong to the Bijawar series and are associated with unmetamorphosed sedimentary rocks. It seems reasonable, therefore, that the basic rocks at Gwalior have not been metamorphosed. 8 oriented samples from a quarry near the Gwalior fort have been analysed.

(e) *Mundwara Complex.*—An assemblage of differentiates (Coulson 1933) comprising alkali syenites, gabbros, basalts etc. is exposed near the village Mundwara in the Sirohi district of Rajasthan. The outcrop is small, ring-shaped, and occurs in the main mass of Erinpura Granites. The age of the Erinpura Granites was determined by Holmes (1949) to be about 750 M.y. 8 oriented samples from the basaltic exposure in the complex have been analysed.

## 3. Experimental procedure

Oriented samples were collected from the available fresh exposures of all the formations mentioned earlier. Each sample was cut into cylindrical specimens $\frac{7}{8}$ inch in diameter and $\frac{3}{4}$ inch in height. An average of 3 specimens was obtained from each sample. The magnetic directions of specimens were initially measured with an astatic magnetometer in Bombay. (The Rhyolites, being weakly magnetized, were measured with more sensitive apparatus at the Institute's field station at Khandala where instruments with gradient sensitivity up to $10^{-7}$ gauss/cm are available.) Representative specimens from samples were then demagnetized in alternating magnetic fields of peak value 50 and 200 Oe and remeasured in both cases. The apparatus used for this was developed by two of us (P.W.S. and C.R.K.) jointly with the group at Imperial College, London, in 1957. The instrument consists of a specimen holder inside a split solenoid, rotating round two perpendicular axes simultaneously with frequencies of rotation in the ratio of 1 : 2. The alternating demagnetizing field is provided by feeding the 50 c/s mains to the solenoid. The whole assembly is kept in a field-free space.

Another set of representative specimens was measured at various temperatures at intervals of 100 °C up to the respective Curie points under a specially constructed magnetometer. The furnace consists of antiparallel platinum wire helices mounted on a mica sheet and rolled into a cylinder inside an asbestos-cement cylinder. The furnace is mounted under the magnetometer and has a double walled pyrex glass cooling jacket to prevent the heat from reaching the magnets of the magnetic system. Temperatures of the order of 700 °C can be easily reached by passing 6 A a.c. from the 50 c/s mains through the heating element which has no effect whatsoever on the magnetometer. Whilst at a desired temperature inside the furnace, the specimen can be rotated through 90° round a horizontal axis to enable measurement of all the three components of the magnetic vector in the specimen. This rotation of specimen is achieved with a quartz or aluminium screwdriver going through the whole exterior of the furnace and engaging in a slot made on the specimen, as described by Wilson (1961).

Only those specimens which gave the same directions both after a.c. demagnetization and at higher temperatures in thermal demagnetization experiments have been used in this analysis. The percentage of such stable specimens was quite high (around 80 per cent or more) in the case of all horizons except (d). If the directions obtained in specimens were markedly different in the above measurements then such specimens were rejected.

It follows, therefore, that unless some unknown mechanism has given the rocks these highly stable remanences, subsequent to their formation, the directions given in Table 1 can be taken to represent the orientation of the prevailing geomagnetic field at various epochs in the past corresponding to formation times of these rocks. It may be mentioned here that the field conditions of these rocks do not indicate that they have been disturbed to any considerable extent since their formation.

## 4. Description of results

In Table 1 are given details of new sampling sites and measurements. The mean intensities of magnetization for these formations are not given in this table since the final directions were computed from measurements on cleaned specimens. Figures 2, 3, 4 and 5 are stereograms for specimens from formations (a), (b), (c) and (e). The stereogram for samples from formation (d) is not included since the directions obtained differed widely from one specimen to another in the same sample and from block to block in the same site. Since demagnetization tests did not improve the situation, the results from formation (d) were omitted from the present discussion.

For the Sylhet traps, two distinct groups of directions are found with almost the same dip but widely different azimuths (Table 1 and Figure 2). The main group has an azimuth of 332° and a dip of 59° up, which agrees closely with that previously obtained for the neighbouring and nearly contemporary Rajmahal traps, where the values are 328° and 64° up. No explanation has been found for the quite different azimuth 243° of the other group. Possibly some local geological movement of some kind, perhaps involving alteration of the rock structure, may have occurred. Further study of these anomalous rocks is needed. In the present analysis this group will not be used.

The measurements for the Malani Rhyolites, the Bijawar traps and the Mundwara complex all give reasonably compact distributions of magnetic directions, (Figures 3, 4 and 5), and the mean directions are widely different from the directions

Stereograms for samples from Sylhet traps (Figure 2), Malani Rhyolites (Figure 3), Bijawar traps (Figure 4) and Mundwara complex (Figure 5).

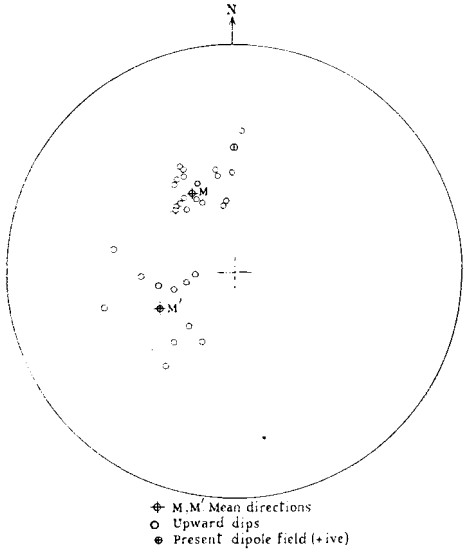



FIG. 2.—Sylhet traps.

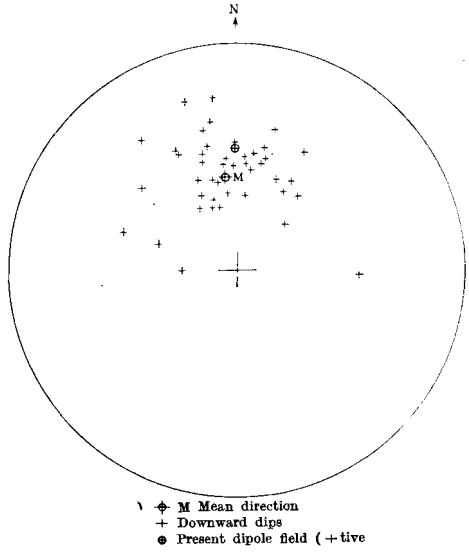



FIG. 3.—Malani Rhyolites.

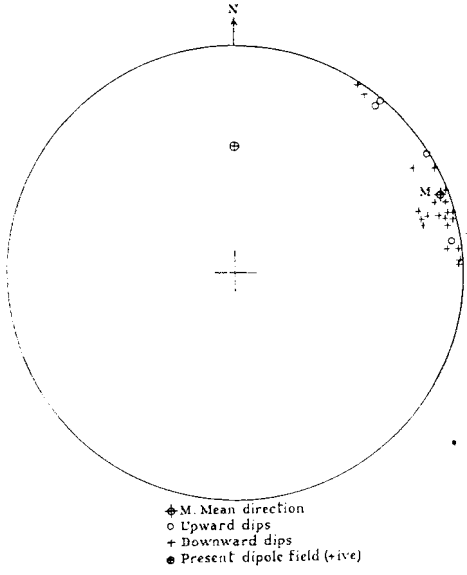



FIG. 4.—Bijawar traps.

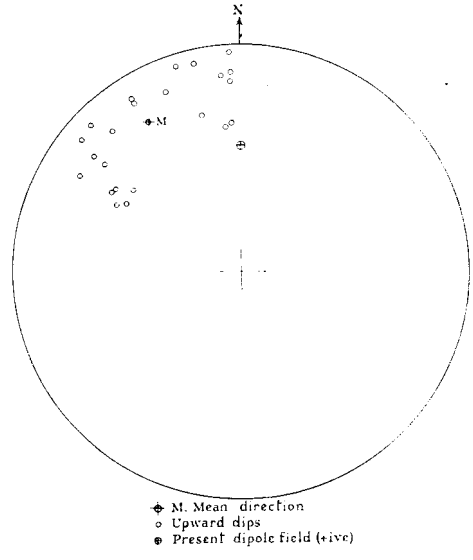



FIG. 5.—Mundwara complex.

of an axial dipole, as is also the case for the Sylhet traps. So a reasonable degree of magnetic stability is indicated for all four formations. Assuming that the field has always been that of an axial dipole, the ancient latitude $\lambda_s$ of the site is calculated from the magnetic dip $I$ by the relation

$$\tan \lambda_s = \tfrac{1}{2} \tan I$$

Then from these latitudes $\lambda_s$ of a site is calculated the latitude $\lambda_m$ of the reference town for the land mass, which for India is taken as Nagpur (Blackett & others 1960). The resulting ancient latitudes of Nagpur are given in column 8 of Table 1.

## Table 1

*New measurements of Indian rocks*

| 1 | 2 | 3 | | 4 | 5 | 6 | | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Formation studied | Estimated age (M.y.) | Mean position of site | | No. of samples | No. of specimens | Mean | | Precision* | Ancient latitude referred to Nagpur |
| | | Lat. | Long. | | | azimuth | dip | | |
| Sylhet traps | 150 | 25N | 91E | $36^{25}_{11}$ | $100^{80}_{20}$ | 332<br>243 | −59<br>−60 | 7°<br>16° | 43 °S<br>43° S |
| Malani Rhyolites | 500 | 26N | 73E | 60 | 170 | 353 | −56 | 10° | 34 °N |
| Bijawar traps | 600 | 26N | 78E | 7 | 25 | 70 | 3 | 18° | 2 °S |
| Cuddapah traps | 600 | 14N | 78E | 72 | 245 | — | — | — | — |
| Mundwara complex | 750 | 25N | 73E | 8 | 25 | 329 | −24 | 21° | 13 °S |

* Radius of circle of 95 per cent confidence (Fisher 1953).
Dip: + down, — up.

In Figures 6(a) and 7(a) are collected all the available results for $\lambda_m$ and the rotation $\psi$ for Indian rocks. The earlier results (Clegg & others 1958; Deutsch & others 1959), are shown by black dots and the new ones decribed in this paper by open circles; apart from the uncertainty arising from the anomalous group of the Ambo rocks (Deutsch & others 1959, p. 46), the mean curves representing the rotation $\psi$ and the calculated latitude $\lambda_m$ of the reference town can be considered fairly well established back to 150 M.y. However, the next point occurs at 500 M.y., leaving a gap of 350 M.y. in which there are no magnetic measurements. It is not therefore possible to exclude the occurrence of big changes in $\psi$ and $I$ during this interval. If the azimuth exceeded 90° during this interval, implying that India had rotated subsequently through more than this angle, then it follows that the sign of the ancient latitude, as determined from the measured dip, must be reversed. In fact the measured azimuth at 600 M.y. is 70°, so a rotation through more than 90° between 500 and 750 M.y. may be suspected: if so, the relative signs of the latitude at 500, 600 and 750 M.y. will be in doubt.

## 5. Comparison of results from four continents

In Figures 6 and 7 are also given the analogous results from Australia, North America and Europe. The curves are the extension to 750 M.y. of those given to 500 M.y. by Blackett & others (1960), making use of the additional data given in Table 2 taken from the compilation by Cox & Doell (1960). Only those results, mostly from igneous rocks, for which the authors give evidence of stability are used in the analysis.

Though the magnetic data at the earlier ages is scanty and of low accuracy, and moreover ambiguities of the sign of the latitudes and orientations may exist, it is of interest to consider the results for the four continents as a whole.

The most striking features of these curves is the well-known common northward movement of the four continents in the last 250 M.y. At earlier times both northward and southward movements occur. Apart from this, there is no very obvious change with time of the average pattern of drift since 750 M.y. From the numerical data in the tables it can be calculated that the average rate of drift of the four continents over the period of 750 M.y. is about 0°·16 per million years or 1·7 cm a year: the corresponding rate of change of the orientations is about 0°·2 per million years.

Assuming as is usual that the drift is due to convection currents in the mantle (Heiskanen & Vening Meiesz 1958; Chandrasekhar 1961; Runcorn 1962), this result would imply that the average pattern of the convection currents has not changed markedly in 750 M.y. However, the paucity of the data makes any reliable correlation impossible. It is very important that the curves of $|\lambda|$ and $|\psi|$ against $T$ be determined much more accurately and at much closer time intervals as for the other continents. Only then would it be possible to eliminate the ambiguities of sign and so to determine $\lambda$ and $\psi$ reliably. When continuous curves of these two variables are available, it will be possible to test whether the rates of change $d\lambda/dt$ and $d\psi/dT$ have changed systematically with age. Then by studying the distribution of the pole positions obtained from the different land masses at different epochs, it will be possible to deduce whether the general pattern of drift observed today held in the distant past, or whether several of the continents were for long periods rigidly locked together (e.g. Gondwanaland) as many have held.

**Table 2**

*New Indian results together with pre-Cambrian results by other workers for N. America, Europe and Australia*

| Rock form | Geological age | Estimated age M.y. | Lat. | Long. | Remanent Decl. E of N | Directn. Incl. | Inclination difference $\Delta I$ | Rotation $\psi$ | Ancient latitude $\lambda m$ | Divergence $\phi$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| *India* | | | | | | | | | | |
| Sylhet traps | Jurassic | 150 | 25N | 91E | 332 | −59 | −102 | −28 | 40S | 92 |
| | | | | | 243 | −60 | | | | |
| Malani Rhyolites | Lower Vindhyan | 500 | 26N | 73E | 353 | 56 | +28 | −7 | 34N | 28 |
| Bijawar traps | Algonkian | 600 | 26N | 78E | 70 | 3 | −41 | 70 | 2S | 73 |
| Mundwara complex | pre-Cambrian | 750 | 25N | 73E | 329 | −24 | −67 | −31 | 13S | 73 |
| *Australia* | | | | | | | | | | |
| Buldiva quartzites | Late pre-Cambrian | 600 | 14S | 132E | 243 | 38 | −12 | −63 | 30S | 54 |
| Nullagine lavas | Late pre-Cambrian | 650 | 21S | 120E | 143 | 64 | −26 | −37 | 48S | 34 |
| Edith river volcanics | Earliest late pre-Cambrian | 700 | 13S | 132E | 90 | 48 | −23 | −90 | 39S | 72 |
| *Europe* | | | | | | | | | | |
| Torridonian Sst | Late pre-Cambrian | 600 | 58N | 6W | 123 | 44 | −117 | −57 | 26S | 47 |
| Longmyndian sed. | | 600 | 58N | 6W | 114 | 29 | −98 | −66 | 15S | 55 |
| L. Torridonian Sst. | | 650 | 53N | 3W | 307 | 34 | −107 | −53 | 19S | 47 |
| *N. America* | | | | | | | | | | |
| Copper harbour lavas sed. | Late Keweenawan | 600 | 47N | 88W | 294 | 32 | −33 | −66 | 17N | 51 |
| Portage lake lavas | Middle Keweenawan | 700 | 65N | 88W | 282 | 41 | −24 | −78 | 23N | 49 |
| Sudbury intrusives | | 1200 | 47N | 81W | 300 | 78 | +13 | −60 | 67N | 22 |

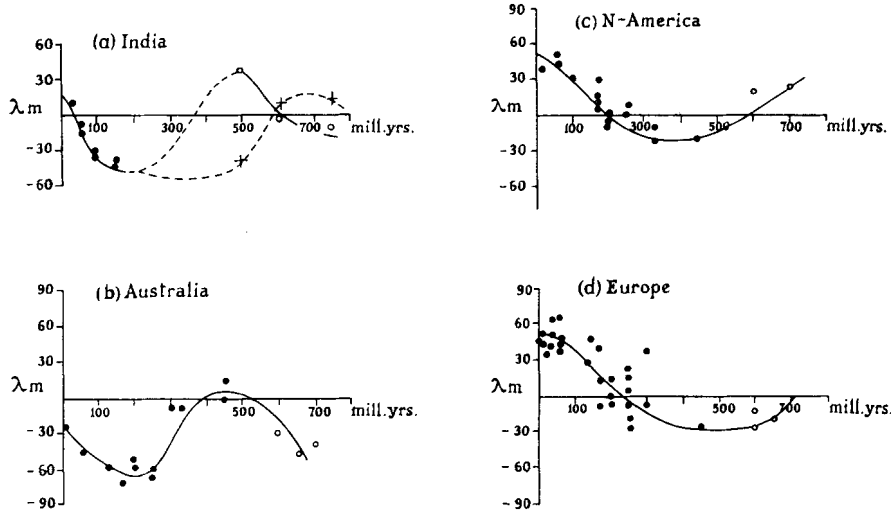



FIG. 6.—Latitudes of four continents. Reference towns: India, Nagpur; Australia, Alice Springs; N. America, Denver; Europe, Paris.

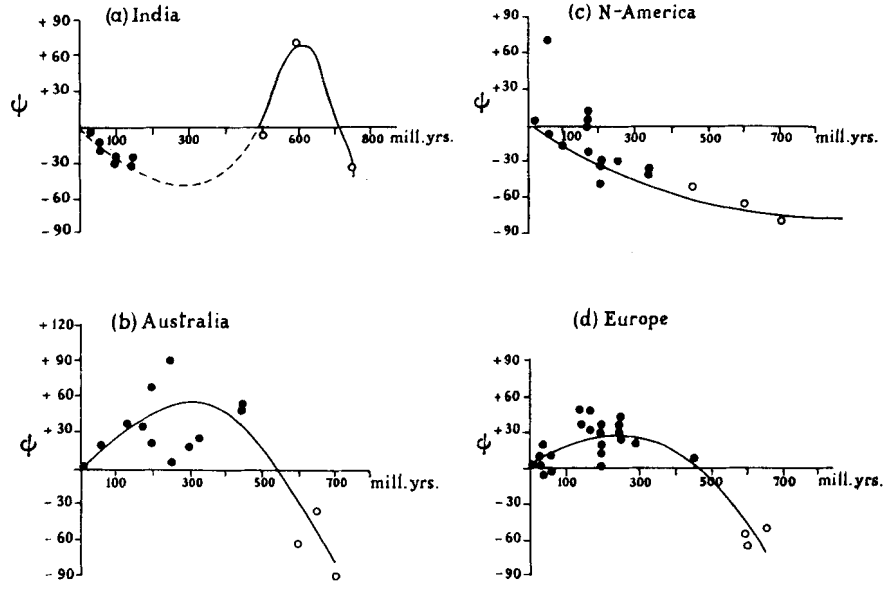



FIG. 7.—Rotations of four continents.

## 6. Acknowledgments

We wish to express our deep sense of gratitude to Dr J. A. Clegg and Professor P. M. S. Blackett for guidance and encouragement throughout this work. We take this opportunity to express our thanks to Professor S. Chandrashekhar for many useful discussions that we had during his recent visit to the Institute. Our thanks are due to Professors M. G. K. Menon and D. Lal for their keen interest and encouragement throughout this work.

It is our pleasure to acknowledge the help we have received from the Geological Survey of India and the Defence Science Organization in planning and carrying out our sampling expeditions.

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1962 *May.*

## References

Blackett, P. M. S., Clegg, J. A. & Stubbs, P. R. S., 1960. *Proc. Roy. Soc. A*, **256,** 291.
Blackett, P. M. S., 1961. *Proc. Roy. Soc. A*, **263,** 1.
Chandrasekhar, S., 1961. *Hydrodynamic and Hydromagnetic Stability*, Clarendon Press, p. 220.
Clegg, J. A., Deutsch, E. R. & Griffiths, D. H., 1956. *Phil. Mag.*, **1,** 419.
Clegg, J. A., Radhakrishnamurty, C. & Sahasrabudhe, P. W., 1958. *Nature (Lond.)*, **181,** 830.
Coulson, A. L., 1933. *Mem. Geol. Surv. Ind.*, **63,** Part I.
Cox, A. & Doell, R. R., 1960. *Bul. Geol. Soc. Amer.*, **71,** 645.
Deutsch, E. R., Radhakrishnamurty, C. & Sahasrabudhe, P. W., 1958. *Phil. Mag.*, **3,** 170.
Deutsch, E. R., Radhakrishnamurty, C. & Sahasrabudhe, P. W., 1959. *Ann. Geophys. (Paris)* **15,** 39.
Dubey, V. S., 1960. Pres. Address, Geology Session, Proc. Indian Scientific Conference.
Fisher, A., 1953. *Proc. Roy. Soc. A.*, **217,** 295.
Heiskanen, W. A., & Vening Meiesz, F. A., 1958. *The Earth and Its Gravity Field*, McGraw Hill Book Co., 397.
Holmes, A., 1949. *Geol. Mag.*, **86,** 288.
Holmes, A., 1960. *Trans. Edin. Geol. Soc.*, **17,** 183.
La Touche, T. D., 1902. *Mem. Geol. Surv. Ind.*, **35,** Part I.
Medlicott, H. B., 1865. *Mem. Geol. Surv. Ind.*, **7,** Part I.
Runcorn, S. K., 1962. *Nature*, **193,** 211.
Wadia, D. N., 1953. *Geology of India*, Macmillan & Co., 118.
Wilson, R. L., 1961. *Geophys. J.* **5,** 46.